

Quantitative Analysis of Prime Numbers Across Base-100 Scales : Linear Divergence of Prime Gaps and the Role of Logarithmic Constant $\ln(10)$

Base 100 reference and preliminary ratios:

- Reference base: 100
- For the first scale (10^1 , corresponding to $1 \leq n \leq 10$), the number of primes is 4, giving an interval-to-prime ratio of $10 / 4 = 2.5$
- Subsequent scales follow the same definition, with each scale representing one order-of-magnitude interval based on the reference base 100

Scale (10^k)	Number of Primes	Interval-to-Prime Ratio	Absolute Increase	Prime Density (%)
10^1	4	2.5	–	40%
10^2	25	4	1.5	25%
10^3	168	5.95	1.95	16.80%
10^4	1,229	8.13	2.18	12.29%
10^5	9,592	10.42	2.29	9.59%
10^6	78,498	12.73	2.31	7.84%
10^7	664,579	15.04	2.31	6.64%
10^8	5,761,455	17.35	2.31	5.76%
10^9	50,847,534	19.66	2.31	5.08%
10^{10}	455,052,511	21.97	2.31	4.55%
10^{11}	4,118,054,813	24.28	2.31	4.11%
10^{12}	37,607,912,018	26.59	2.31	3.76%
10^{13}	346,065,536,839	28.89	2.30	3.46%
10^{14}	3,204,941,750,802	31.20	2.31	3.20%
10^{15}	29,844,570,422,669	33.50	2.30	2.98%
10^{16}	279,238,341,033,925	35.81	2.31	2.79%
10^{17}	2,623,557,157,654,230	38.11	2.30	2.62%
10^{18}	24,739,954,287,740,800	40.42	2.31	2.47%
10^{19}	234,057,667,276,344,000	42.72	2.30	2.34%
10^{20}	2,220,819,602,560,910,000	45.02	2.30	2.22%
10^{21}	21,127,269,486,018,700,000	47.33	2.31	2.11%
10^{22}	201,467,286,689,315,000,000	49.63	2.30	2.01%
10^{23}	1,925,320,391,606,800,000,000	51.93	2.30	1.92%
10^{24}	18,435,599,767,349,200,000,000	54.24	2.31	1.84%
10^{25}	176,846,309,399,143,000,000,000	56.54	2.30	1.76%